P-S-585a
December 12, 1962

SUPERSEDING
Int. Fed. Spec. P-S-00585 (Army-QMC)
January 15, 1962,
Fed. Spec. P-S-563b
April 2, 1958, and
Fed. Spec. P-S-578
August 11, 1955

FEDERAL SPECIFICATION

SOAP, LAUNDRY, BUILT, POWDERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of powdered, built, laundry soaps (see 6.1).

1.2 Classification.

1.2.1 *Types.* Soaps shall be of the following types as specified (see 6.2).

Type I — Low-titer soap

Type II — High-titer soap

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378—Plastic Film, (Polyethylene, Thin Gauge).

UU-S-48—Sacks, Shipping, Paper.

PPP-D-723—Drums, Fiber.

PPP-T-0060—Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid shall apply.

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Composition.** The composition of types I and II soaps shall conform to the requirements specified in table I, when tested as specified in table II.

3.1.1 *Computation.* The percentage of matter volatile at 105°C. shall be calculated on the basis of the soap as received, but all other constituents shall be calculated on the basis of material containing 16.0 percent of matter volatile at 105°C. (for basis of purchase see 6.4). Deliveries containing more than 16 percent of matter volatile at 105°C. shall be rejected.

3.2 **Color.** Unless otherwise specified, the soap shall have a uniform, light color, and shall be no darker than Color Number 30266 in Fed. Std. No. 595 (see 6.3).

3.3 **Odor.** The odor shall not be objectionable in the soap as received.

3.4 **Form.**

3.4.1 *Type I.* The soap shall be a uniform mixture of soap and alkalies in powdered form.

3.4.2 *Type II.* The soap shall be a uniform mixture of soap and alkalies in powdered form.

TABLE I—*Composition.*

Characteristic	Type I		Type II	
	Minimum *Percent*	Maximum *Percent*	Minimum *Percent*	Maximum *Percent*
Moisture and matter volatile at 105°C.	—	16.0	—	16.0
Sum of free alkali and total matter insoluble in alcohol (alkaline salts)	18.5	40.0	17.0	40.0
Matter insoluble in water	—	1.0	—	1.0
Titer of mixed fatty acids prepared from the soap	—	29°C.	39°C.	—
Matter retained in No. 12 sieve	—	1.5	—	1.5
Matter passing a No. 140 sieve	—	20.0	—	20.0
Anhydrous soap content	50.0	—	48.0	—
Free alkali calculated as NaOH	—	0.2	—	16.0
Rosin	—	0.0	—	0.0

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3.5 **Solubility.**

3.5.1 *Type I.* Type I soap shall be readily soluble in water at 100°F. when tested as specified in table II.

3.5.2 *Type II.* Type II soap shall be readily soluble in water at 140°F. when tested as specified in table II.

3.6 **Workmanship.** The finished product shall be clean, uniform, and free flowing.

3.7 **Quality.** The quality shall conform to levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQLs) specified in section 4 for all examinations and tests of materials, components and end item, as applicable.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection and acceptance.** Sampling for inspection and test shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for inspection and tests shall consist of all material of the same type and batch submitted for delivery at one time.

4.3 **Inspection.**

4.3.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 *Testing and inspection of the end item.*

4.3.2.1 *Testing of the end item.* The end item shall be tested for the characteristics specified in table I in accordance with the test methods specified in table II. When four or more containers comprise the shipping lot, a sample of not less than 1 pound shall be taken from each of not less than 4 separate shipping containers, randomly selected throughout the lot presented for examination. When less than 4 shipping containers comprise the lot the total samples shall, in all cases, consist of not less than 3 sample portions taken at random from the containers or container, and shall be placed in a suitable size receptacle. After thorough mixing, a 2 pound composite sample shall be drawn for testing purposes. The composite sample shall be placed in an air-tight container, sealed, and marked with the identifying lot number and date. The average of 2 determinations shall comply with the requirements when numerical results are specified. Both determinations shall pass when "pass" or "fail" requirements are applicable.

4.3.2.2 *Inspection of the end item.* The end item shall be examined for appearance, net contents, markings and preparation for delivery in accordance with the classification of defects and acceptable quality levels (AQ Ls) set forth below. The lot size for determining the sample size shall be expressed in units of filled containers of the same capacity consisting of soap manufactured in one batch and offered for delivery at one

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TABLE II.—*Instructions for testing*

Characteristic	Specification reference		Requirements applicable to		Number determinations on composite sample [2]	Results reported as	
	Requirement	Test method[1]	Individ unit	Lot aver		Pass or fail	Numerically to nearest
Moisture and matter volatile at 105°C.	Table I	201.1			Average of 2		0.1 percent
Sum of free alkali and total matter insoluble in alcohol (alkaline salts)	Table I	301.3			Average of 2		0.1 percent
Matter insoluble in water	Table I	303.1			Average of 2		0.1 percent
Titers of mixed fatty acids prepared from the soap	Table I	901			Average of 2		0.1°C
Matter retained on a number 12 sieve	Table I	2101			Average of 2		0.1 percent
Matter passing a number 140 sieve	Table I	2101			Average of 2		0.1 percent
Rosin	Table I	701.3	X		2	X	
Anhydrous soap content	Table I	401.3			Average of 2		0.1 percent
Free alkali calculated at NaOH	Table I	302.3			Average of 2		0.1 percent
Solubility in water	3.5.1 3.5.2	4.4.1	X		2	X	

[1] All tests except solubility are performed in accordance with Federal Test Method Standard No. 536.
[2] If failure is indicated, report description of failure or numerical point of failure as applicable.

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time for the examinations in 4.3.2.2.1, 4.3.2.2.2 and 4.3.2.2.3. The inspection levels for purposes of determining the sample size in accordance with the provisions of MIL-STD-105 and the acceptable quality levels (AQL's) shall be as specified in 4.3.2.4.

4.3.2.2.1 *Examination of the primary container for defects in closure, markings, appearance, construction, and preparation for delivery.* The sample unit for this examination shall be one filled primary container fully prepared for shipment.

Examine	*Defect*
Construction	Any seepage of contents. Seams or closures not as specified. Not in container specified.
Workmanship	Any tear, hole, break or scuff in container.
Markings	Omitted; illegible; incomplete; smeared; incorrect; not as specified.

4.3.2.2.2 *Examination of the end item for defects in appearance, odor, and form.* The sample unit for this examination shall be the contents of one filled primary container.

Examine	*Defect*
Color	Too dark. Darker than Color Chip Number 30266.
Odor	Objectionable.
Form	Not uniform powder, contains hard lumps. Not free flowing.

4.3.2.2.3 *Examination of the end item for defects in net contents.* The sample unit for this examination shall be one filled container. The average net weight per container shall be not less than specified or indicated.

4.3.2.2.4 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

	Inspection level	*AQL*
Examination paragraph 4.3.2.2.1	I	1.5
Examination paragraph 4.3.2.2.2	L-4	2.5
Examination paragraph 4.3.2.2.3	L-4	—

4.4 **Test procedures.**

4.4.1 *Solubility in water.* Add 0.5 grams of soap (type I or type II) to 100 grams of water 100°±2°F. or 140°±2°F., as applicable. After moderate stirring for ten minutes the soap solution shall be clear and free from sediment.

5. PREPARATION FOR DELIVERY.

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurement.)

5.1 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A.* The laundry powder shall be packed in 50, 100, or 300-lb. quantities as specified (see 6.2). Final closure of each shipping container shall not be completed until the temperature of the soap contents has cooled to room temperature.

5.1.1.1 *Fifty-pound quantity.* Fifty pounds of one type soap shall be packed in a multiwall paper shipping sack conforming to construction number 2X of UU-S-48.

5.1.1.2 *One-hundred and three-hundred-pound quantities.* One hundred or three hundred pounds of one type soap shall be packed in a fiber drum conforming to type III, grade A of PPP-D-723. Closure of the drum shall be effected by covering the joint

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between the cover and body by one complete turn of minimum 3-inch width tape conforming to type III, class 1 of PPP-T-0060. The tape shall overlap itself a minimum of two inches. Each drum shall be lined with a minimum 0.002-inch thick polyethylene bag conforming to type I, grade A, finish 1 of L-P-378. All seams and closures shall be effected by heat sealing and shall conform to the heat seal strength requirement in L-P-378.

5.1.2 *Level B.* The laundry powder shall be packed in 50-, 100-, or 300-lb. quantities as specified (see 6.2). Final closure of each shipping container shall not be completed until the temperature of the soap contents has cooled to room temperature.

5.1.2.1 *Fifty-pound quantity.* Fifty pounds of one type soap shall be packed in a multiwall paper sack conforming to construction number 1 L/W of UU-S-48.

5.1.2.2 *One-hundred and three-hundred-pound quantities.* One hundred or three hundred pounds of one type soap shall be packed in a fiber drum conforming to type II, grade A of PPP-D-723. Each drum shall be closed and lined as specified in 5.1.1.2.

5.1.3 *Level C.* Soap powder of one type shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. In addition, each shipping container shall be marked "STORE IN A COOL DRY PLACE".

6. NOTES

6.1 Intended use. Type I soap is intended for low temperature laundering and general cleaning where the presence of alkaline soaps is desirable. It is not intended for use on silk, woolen, or synthetic fabrics.

Type II soap is intended for high temperature laundering of cotton and linen fabrics. It is not intended for use on silk, woolen or synthetic fabrics.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required.

(*c*) Inspection responsibility, methods, and tests (section 4).

(*d*) Selection of applicable levels of packing (5.1).

(e) Weight of pack required (5.1.1 and 5.1.2).

6.3 Color chips. Color chips listed in Fed. Std. No. 595 may be purchased at five cents each from any Business Service Center, of General Services Administration.

6.4 Basis for purchase. The soap shall be purchased by net weight provided the matter volatile at 105° C. is neither above or below 13 percent. Deliveries containing more than 16 percent of matter volatile shall be rejected without further test. On deliveries containing 16 percent or less of matter volatile at 105° C. payment should be made on a basis of a product containing 13 percent moisture. The net weight basis is calculated as follows:

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$$W = \frac{R \times (100\text{-}L)}{87}$$

Where: W = net weight to be paid for on 13 percent moisture and matter volatile basis.

R = net weight of material as received.

L = percentage loss at 105° C.

6.5 Transportation description. Transportation description applicable to this item is:

Soap, not otherwise indexed by name, other than liquid.

Carload minimum weight 36,000 pounds. Truckload minimum weight 36,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Q

Navy—Sh

Air Force—MOA

OTHER INTERESTS:

Army—M

Navy—S MC

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